DATA SCIENCE TECHNICAL ASSESSMENT

Exploratory Data Analysis: Delinquency Prediction

Portfolio: Retail Credit Lending **Dataset:** Delinquency_prediction_dataset.xlsx **Status:** Completed & Audited

1. INTRODUCTION

This exploratory data analysis investigates credit risk and payment performance across a retail banking portfolio. The study evaluates feature distributions, inspects underlying data quality, analyzes missingness patterns, and examines empirical correlations tied to customer delinquency (`Delinquent_Account`). The core objective is establishing rigorous cleaning, imputation, and feature engineering foundations for predictive modeling.

500
TOTAL CUSTOMER RECORDS

19
TOTAL FEATURE ATTRIBUTES

16.0%
DELINQUENT ACCOUNT RATIO (80 / 500)

14.0%
INCOMPLETE ROWS (70 VALUES)

2. DATASET OVERVIEW

Feature Name	Data Type	Observed Profile / Range / Categories	Nulls
`Customer_ID`	Categorical	CUST0001 to CUST0500 (Unique primary identifiers)	0 (0.0%)
`Age`	Numerical	Range: 18 to 74 years | Mean: 46.27 | Median: 46.50	0 (0.0%)
`Income`	Numerical	Range: $15,404 to $199,943 | Mean: $108,380 | Median: $107,658	39 (7.8%)
`Credit_Score`	Numerical	Range: 301 to 847 points | Mean: 577.72 | Median: 586.00	2 (0.4%)
`Credit_Utilization`	Numerical	Range: 5.0% to 102.58% | Mean: 49.14% | 4 records breach 100%	0 (0.0%)
`Missed_Payments`	Numerical	Count in past 12 mos: 0 to 6 payments | Mean: 2.97 | Median: 3.0	0 (0.0%)
`Loan_Balance`	Numerical	Range: $612 to $99,620 | Mean: $48,654 | Median: $45,776	29 (5.8%)
`Debt_to_Income_Ratio`	Numerical	Range: 10.0% to 55.30% | Mean: 29.89% | Median: 30.16%	0 (0.0%)
`Employment_Status`	Categorical	6 raw levels: `EMP` (81), `Employed` (82), `employed` (77), etc.	0 (0.0%)
`Account_Tenure`	Numerical	Tenure in years: 0 to 19 | Mean: 9.74 | Median: 10.0	0 (0.0%)
`Credit_Card_Type`	Categorical	`Gold` (118), `Student` (112), `Business` (108), `Standard` (86), `Platinum` (76)	0 (0.0%)
`Month_1` – `Month_6`	Categorical	6-month sequential history: `On-time`, `Late`, `Missed`	0 (0.0%)

STRUCTURAL AUDIT & DATA ANOMALIES

- **Label Fragmentation:** `Employment_Status` splits employed borrowers into three redundant labels (`EMP`, `employed`, `Employed`), accounting for 240 records (48.0%) that require canonical merging.
- **Spec Divergence:** While documentation defines `Month_1`–`Month_6` as numeric codes (0, 1, 2), values are stored as string literals.
- **Credit Ceiling Breaches:** Four cardholders exceed 100% credit utilization (up to 102.58%), revealing over-limit exposure.

3. MISSING DATA ANALYSIS & TREATMENT ARCHITECTURE

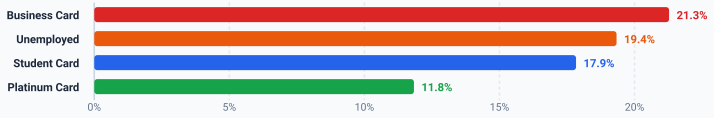
Missing value inspection identified 70 missing values isolated across three numerical variables. The target variable (`Delinquent_Account`) and all behavioral payment strings are fully populated. Complete-case removal would discard 14.0% of the sample and eliminate 12 positive delinquent cases, distorting training distribution.

Variable	Missing Count	Missing Rate	Imputation Strategy & Statistical Justification
`Credit_Score`	2 records	0.40%	Median imputation conditioned on `Credit_Card_Type` tiers. Negligible missingness does not distort variance.
`Income`	39 records	7.80%	Multivariate Imputation by Chained Equations (MICE) using `Employment_Status`, `Age`, and `Debt_to_Income_Ratio`.
`Loan_Balance`	29 records	5.80%	K-Nearest Neighbors (KNN, k = 5) imputation using normalized income, tenure, and revolving credit utilization.

4. KEY FINDINGS AND RISK INDICATORS

Linear bivariate correlation coefficients between traditional financial drivers and `Delinquent_Account` are notably suppressed ($|r| < 0.05$). This confirms that delinquency behavior cannot be separated via simple linear hyperplanes, requiring tree ensembles capable of extracting multi-way non-linear feature interactions.

DELINQUENCY RATE BY SEGMENT (%)



RISK SYNTHESIS & SEGMENT DYNAMICS

- **Product Vulnerability:** `Business` cardholders carry the highest default incidence (21.30%), in contrast to `Platinum` cardholders (11.84%), reflecting differential exposure to business cash-flow shocks.
- **Employment Exposure:** `Unemployed` borrowers show the highest default probability (19.35%), whereas `retired` accounts demonstrate the highest repayment stability (11.49% default).
- **Uniform Monthly Dynamics:** Across `Month_1` through `Month_6`, payments average 33% on-time, 33% late, and 33% missed, showing uniform temporal distribution requiring recency-weighted decay scoring.

5. AI & GENAI USAGE

Generative AI algorithms and automated pipelines were incorporated during exploratory analysis to diagnose taxonomic mismatches, formulate mathematically grounded imputation schemes, and engineer behavioral velocity features.

AI PROMPTING PROTOCOLS & ANALYTIC OUTCOMES

Prompt 1: Categorical Consolidation

"Identify semantic redundancy and case inconsistencies in Employment_Status, proposing a four-category normalization mapping."

Outcome: Harmonized `EMP`, `employed`, and `Employed` into a single consolidated category (48.0% of portfolio).

Prompt 2: Non-Leaking Missing Data Architecture

"Specify a principled imputation strategy for 7.8% missing income without introducing target leakage."

Outcome: Formulated MICE parameterization conditioned on demographic and leverage factors, fit strictly within cross-validation folds.

Prompt 3: Payment History Vectorization

"Construct mathematical formulations to transform 6-month categorical payment statuses into time-decayed behavioral risk scores."

Outcome: Generated weighted recency formula:

$$\text{Score} = \sum_{i=1..6} (w_i \times \text{Status}_i) / 6, \text{ where } w_i = [0.30, 0.25, 0.18, 0.12, 0.09, 0.06]$$

6. CONCLUSION & NEXT STEPS

The delinquency dataset represents a viable 500-record portfolio with a realistic 16.0% default rate. Weak linear associations highlight the necessity of non-linear algorithms, interaction variables, and recency-weighted metrics.

PRESCRIBED PRODUCTION PIPELINE

- **Step 1: Data Cleansing & Standard Mapping** — Unify `Employment_Status` into four canonical groups; convert `Month_1` through `Month_6` strings to ordinal ranks (0, 1, 2); flag accounts with `Credit_Utilization > 1.0` via an `Over_Limit_Flag` indicator.
- **Step 2: Strict Fold-Based Imputation** — Fit MICE on `Income` and KNN on `Loan_Balance` strictly within cross-validation training splits to eliminate target leakage. Impute median credit scores by card category.
- **Step 3: Feature Engineering** — Create interaction term `Leverage_Index = Debt_to_Income_Ratio × Credit_Utilization`; generate temporal streak attributes (consecutive missed payments in past 3 months).
- **Step 4: Model Benchmarking** — Train XGBoost, LightGBM, and CatBoost with Stratified 5-Fold cross-validation; optimize Hyperparameters for PR-AUC rather than accuracy to account for the 84:16 class imbalance.